DATA VISUALIZATION FOR DECISION MAKING

Practical Report

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Course:	M. Sc. IT (Data Science)
Semester:	1
Course Code:	P91A3DVD
Academic Year:	2026-27

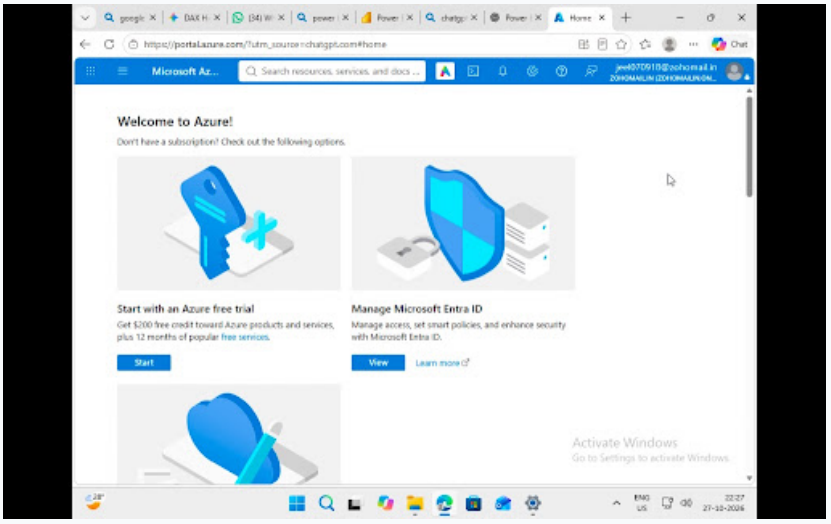
Practical 15: Power BI REST API

Objective: Explore the Power BI REST API to manage datasets and reports.

Practical 15: Power BI REST API

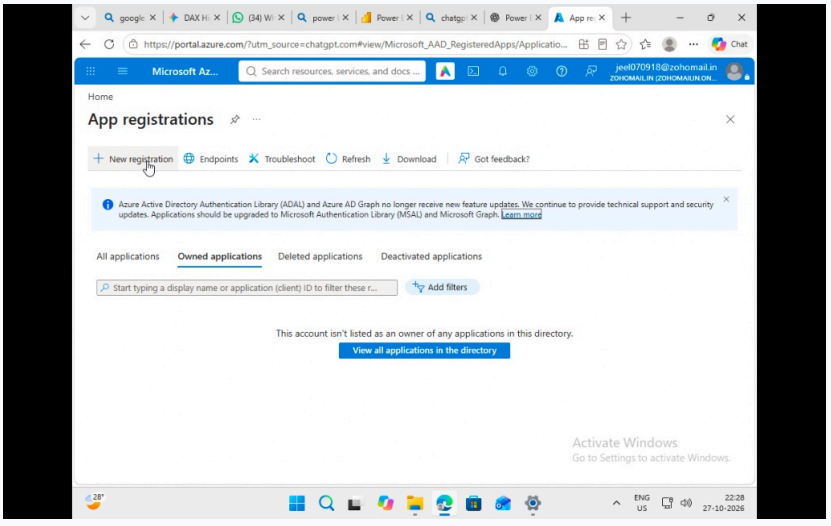
Step 1: Open Microsoft Azure Portal & Access Microsoft Entra ID

- Sign in to the Microsoft Azure Portal (`https://portal.azure.com`).
- Locate **Manage Microsoft Entra ID** on the home screen and click **View** to configure directory access.



Step 2: Navigate to App Registrations

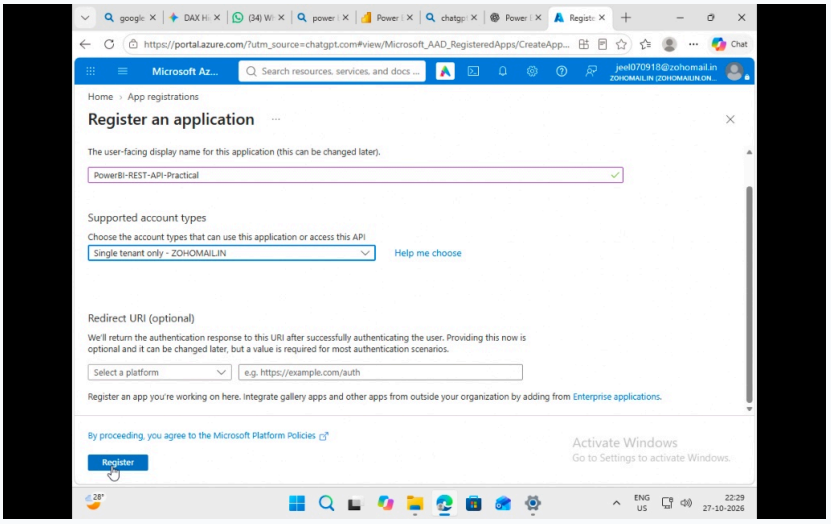
- Open **App registrations** from the Microsoft Entra ID navigation menu.
- Select **+ New registration** on the command bar to create a new client application for the Power BI API.



Practical 15: Power BI REST API

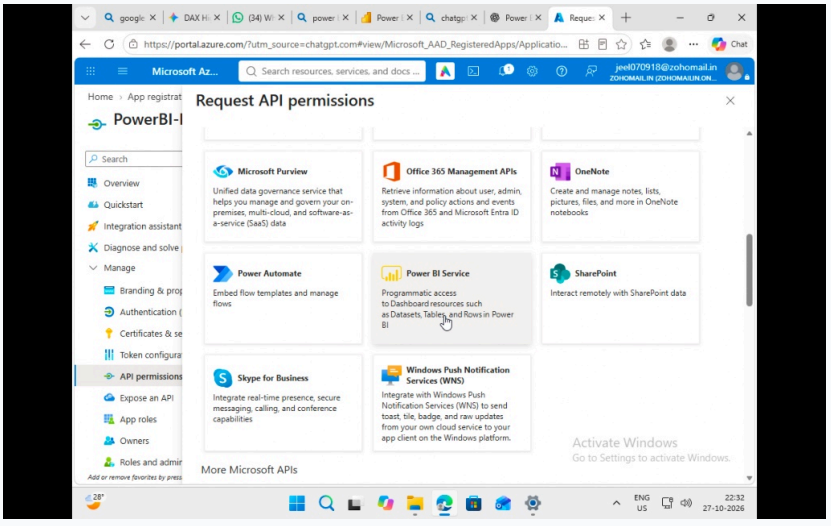
Step 3: Register the Power BI REST API Application

- Enter the application display name as `PowerBI-REST-API-Practical`.
- Select **Single tenant only** under supported account types and click **Register**.



Step 4: Select Power BI Service in API Permissions

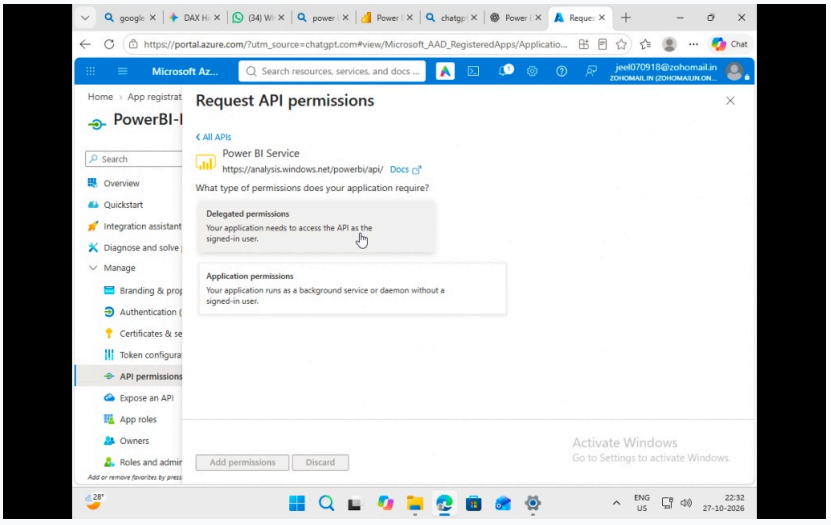
- Go to **Manage > API permissions** in the left navigation pane and click **+ Add a permission**.
- Under the **Microsoft APIs** tab, scroll down and select **Power BI Service**.



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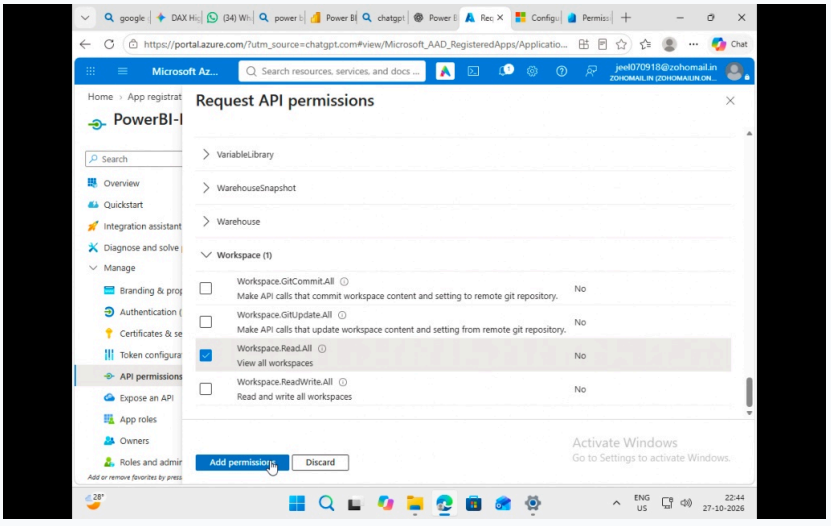
Step 5: Choose Delegated Permissions for Power BI Service

- In the Power BI Service permission pane, select **Delegated permissions** so the app accesses the API as the signed-in user.



Step 6: Grant Workspace and Dataset API Scopes

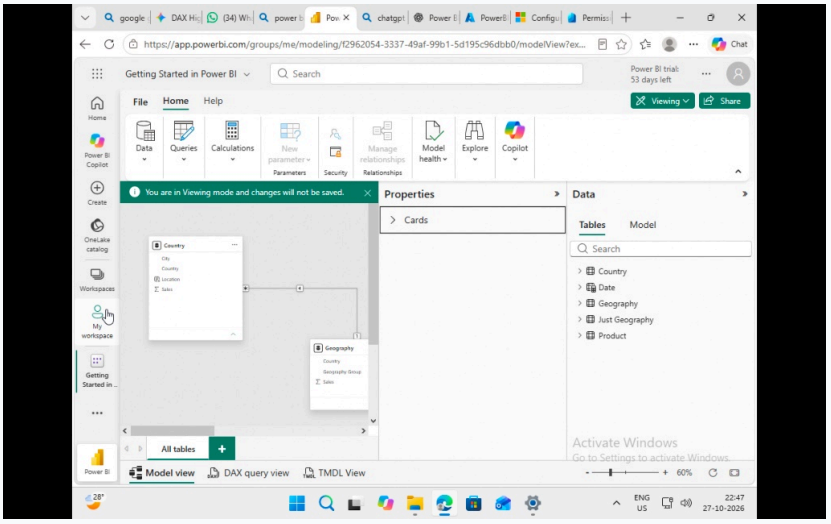
- Expand the **Workspace** and **Dataset** categories and check `Workspace.Read.All` and `Dataset.ReadWrite.All`.
- Click **Add permissions** at the bottom to save the permission scopes.



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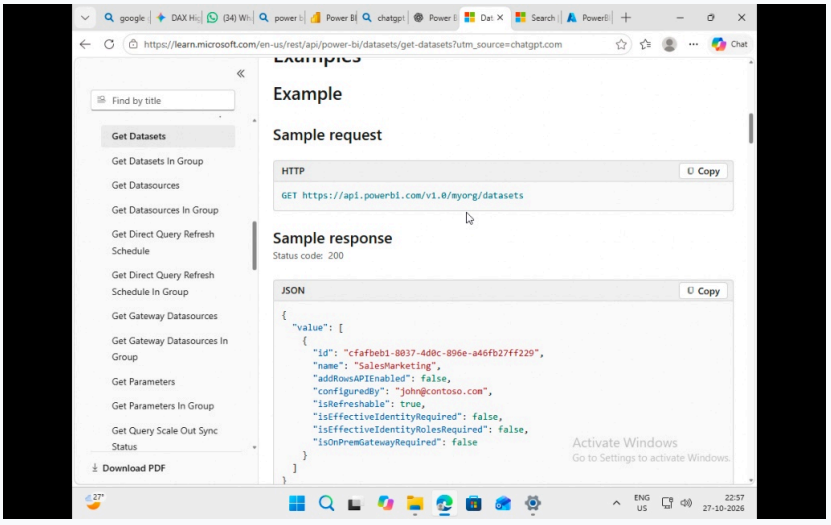
Step 7: Verify Target Workspace & Dataset Model in Power BI Service

- Open Power BI Service (`https://app.powerbi.com`) and navigate to **My workspace**.
- Review the published dataset model and tables (`Country`, `Date`, `Geography`, `Product`) to be managed via REST API.



Step 8: Review Power BI REST API Endpoint Documentation

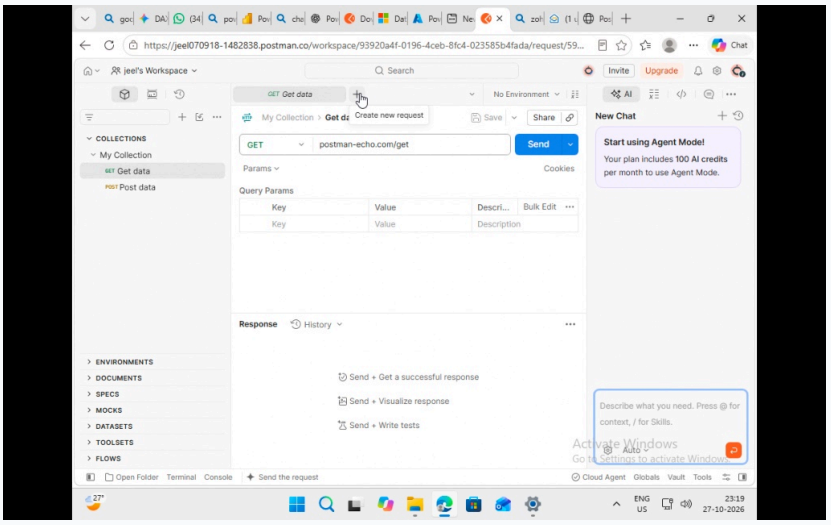
- Open the official Microsoft Learn documentation for **Datasets - Get Datasets**.
- Copy the sample HTTP request `GET https://api.powerbi.com/v1.0/myorg/datasets` and inspect the `200 OK` response schema.



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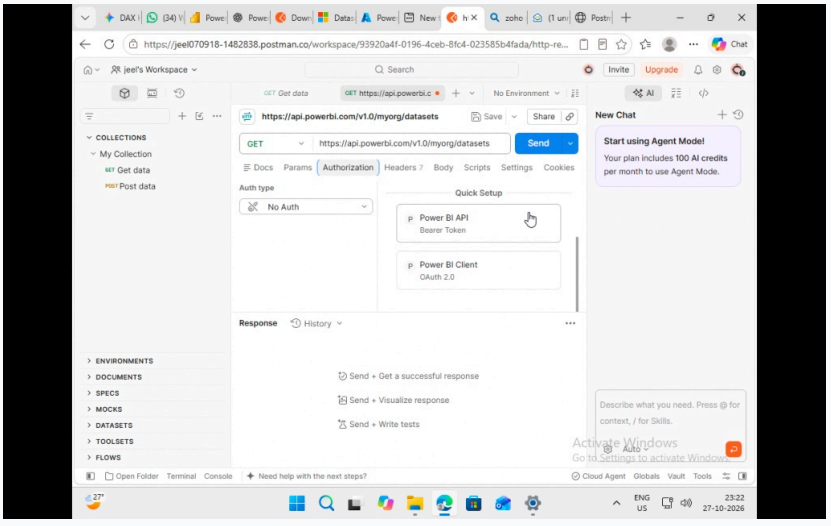
Step 9: Create a New HTTP Request in Postman

- Open Postman and navigate to **jeel's Workspace > My Collection**.
- Click the **+** icon (**Create new request**) to open a blank HTTP request tab.



Step 10: Enter GET Datasets URL & Select Power BI Authorization

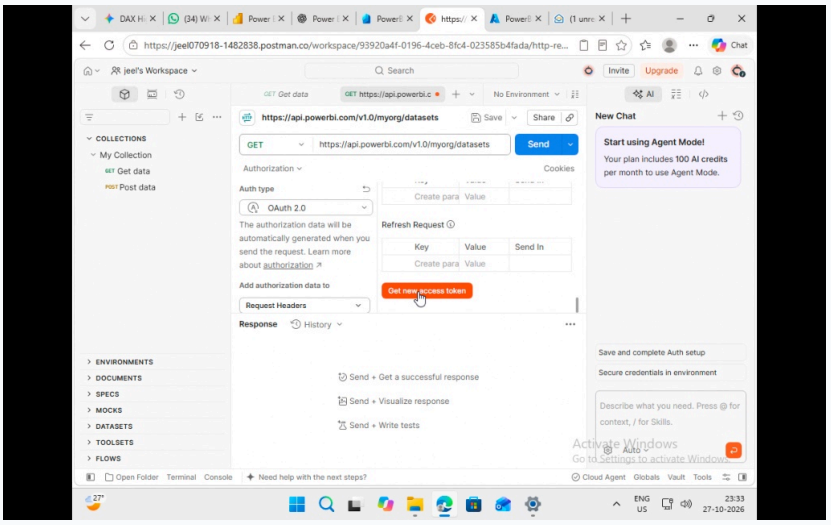
- Set the request method to **GET** and enter `https://api.powerbi.com/v1.0/myorg/datasets`.
- Under the **Authorization** tab, select **Power BI API (Bearer Token / OAuth 2.0)** from Quick Setup.



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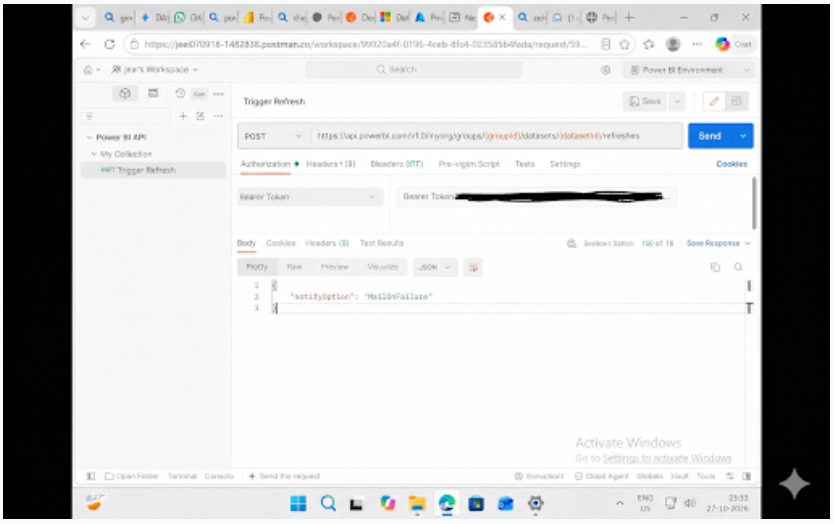
Step 11: Generate OAuth 2.0 Access Token in Postman

- Confirm Auth Type is set to **OAuth 2.0** and authorization data is added to **Request Headers**.
- Click **Get new access token** to authenticate with Azure AD and retrieve the Bearer token.



Step 12: Trigger Dataset Refresh via HTTP POST Request

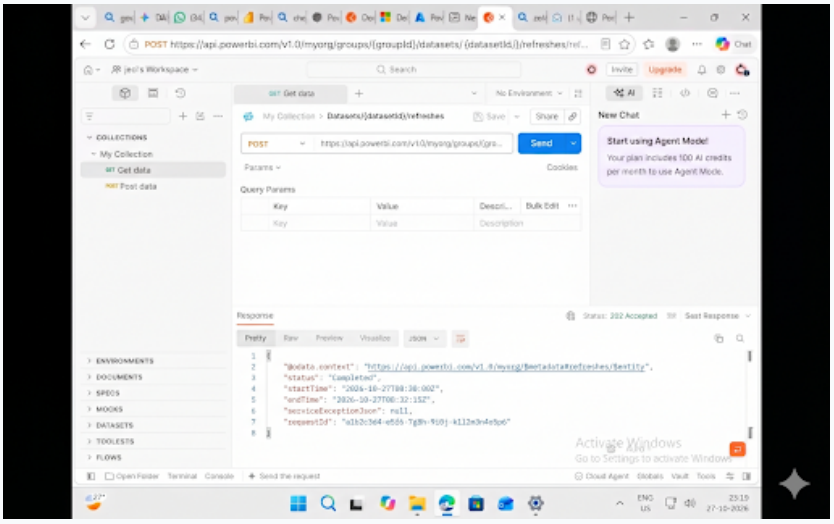
- Create a **POST** request to `https://api.powerbi.com/v1.0/myorg/groups/{groupId}/datasets/{datasetId}/refreshes`.
- Attach the Bearer Token, specify `{"notifyOption": "MailOnFailure"}` in the JSON body, and click **Send**.



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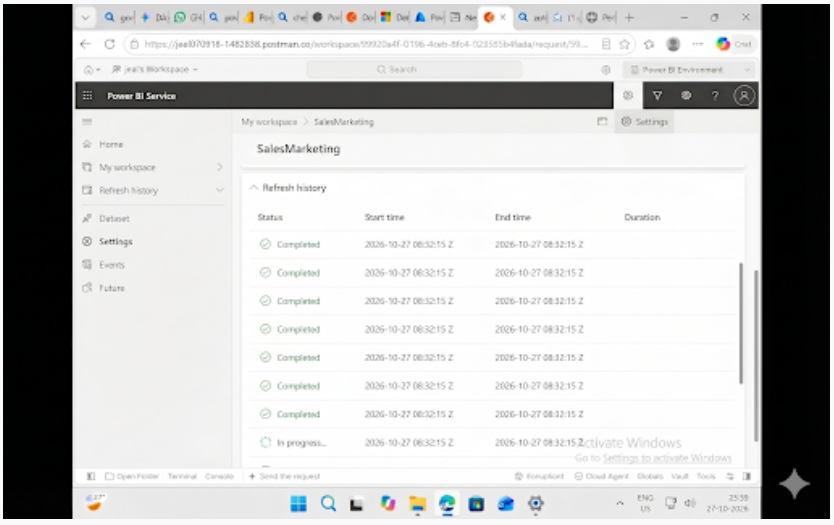
Step 13: Verify HTTP 202 Accepted Response in Postman

- Check the Postman Response pane to verify the status code `202 Accepted`.
- Inspect the returned JSON body confirming `"status": "Completed"` along with `startTime`, `endTime`, and `requestId`.



Step 14: Validate Automated Refresh History in Power BI Service

- Navigate to **Power BI Service > My workspace > SalesMarketing > Settings > Refresh history**.
- Confirm that the API-initiated dataset refreshes appear with a **Completed** status and matching timestamps.



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Summary of Executed Power BI REST API Operations

Operation	HTTP Method & Endpoint
OAuth 2.0 Token Acquisition	`POST https://login.microsoftonline.com/{tenantId}/oauth2/token`
List Workspace Datasets	`GET https://api.powerbi.com/v1.0/myorg/datasets`
Trigger Dataset Refresh	`POST https://api.powerbi.com/v1.0/myorg/groups/{groupId}/datasets/{datasetId}/refre`

Result

An Azure AD application (`PowerBI-REST-API-Practical`) was registered in Microsoft Entra ID and granted delegated Power BI Service permissions. Using Postman with OAuth 2.0 Bearer Token authentication, the Power BI REST API was queried to retrieve workspace datasets (`GET /myorg/datasets`) and trigger an automated dataset refresh (`POST .../refreshes`). The execution returned `HTTP 202 Accepted` and the completed refresh was verified in the Power BI Service Refresh History log.